Intern Management System (IMS) Core Architecture & Workflow Engine

1. Executive Summary

The Intern Management System (IMS) addresses a critical gap in organizational tooling: the disconnect between technical workflow execution and human resources compliance. Current solutions force a compromise. Engineering teams use technical task trackers (Jira, Linear) that are blind to human learning velocity, while HR teams mandate subjective, disconnected survey platforms (Lattice, Workday) for performance reviews.

The core innovation of the IMS is the **Translation Layer**. By deploying a developer-first Kanban workflow engine, the system passively records state transitions, time-in-column metrics, and mentor interventions. These technical actions are systematically translated into actionable, timestamped HR evaluation metrics without requiring dual data entry.

2. Core Architectural Pillars

Agnostic State Machine

Unlike rigid Scrum tools, the IMS allows departments to define custom Kanban columns (e.g., "Awaiting MD Approval" vs. "UAT Testing"). The backend tracks transition deltas, enabling universal performance calculations independent of department nomenclature.

Append-Only Ledger

Task status is not updated in place. Every drag-and-drop action appends a record to an immutable transition ledger, capturing timestamps, previous states, and actor IDs to satisfy strict enterprise audit and compliance requirements.

Threshold Intercepts

Admins map optional SLA thresholds to specific board columns. Background chron workers evaluate the transition ledger against these thresholds, triggering neutral, automated alerts when tasks stagnate, isolating blockages in real-time.

Skill & Friction Tagging

Tasks are linked to specific competencies (e.g., #Python, #Copywriting). When an anomaly occurs (e.g., three revision cycles), the system captures structured tags, proving exactly where an intern is succeeding or struggling.

3. The State-Transition Engine

The foundation of the IMS metrics relies on tracking friction rather than billable output. Time-to-delivery is heavily influenced by external factors, making raw velocity an unreliable metric for junior talent. Instead, the IMS isolates *Revision Cycles* and *Blocked Time*.

When an intern moves a task across the board, the system calculates the time elapsed since the previous transition. If a task moves backward against the designated workflow (e.g., from "Review" to "In Progress"), the system increments the task's Revision Cycle count. This quantitative data protects both the intern (proving a task was stalled by a third party) and the enterprise (documenting a lack of technical progression).

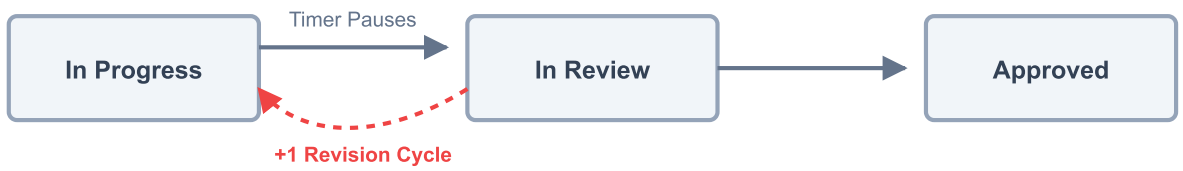


Fig 1. Identifying friction: Backward transitions automatically flag a Revision Cycle for HR review.

> **The Developer Benefit:** Mentors and engineering leads do not interact with timers, spreadsheets, or stopwatches. Their only interface is the Kanban board. The metrics are generated as a passive byproduct of natural workflow execution.

4. The "Justify this Action" Intercept

Raw transition data identifies *where* a blockage occurred, but it cannot determine *why*. To bridge the gap to human resources, the IMS utilizes threshold intercepts. If an intern's task exceeds a defined column threshold or triggers a repetitive revision cycle, the UI intercepts the mentor's drag-and-drop action.

The mentor is presented with a modal requiring them to categorize the friction from a pre-defined HR matrix (e.g., Knowledge Gap, Careless Oversight, Changing Scope, Third-Party Blocker). This micro-interaction occurs in real-time, preventing the common failure mode of relying on a manager's memory at the end of a 90-day internship lifecycle.

HR DASHBOARD AGGREGATION

When the cohort reaches a milestone (30, 60, or 90 days), the system aggregates these intercepts alongside the ledger metrics, auto-populating the performance review dossier.

Skill / Competency	Total Tasks	Avg. Revision Loops	Primary Intercept Tags	System Assessment
`#Python` (Backend)	14	2.8 (High)	Knowledge Gap (4x)	Requires immediate pairing.
`#React` (Frontend)	22	1.1 (Standard)	Scope Creep (2x)	Performing above expectations.
`#Communication`	N/A	N/A	Careless Mistakes (3x)	Rushing deliveries; warn in 1:1.
`#Docker` (Infra)	4	0.5 (Low)	None	Proficient; assign harder tasks.

> **Enterprise Value Proposition:** By automatically generating this structured evidence over the course of the internship, HR is provided with an undeniable, data-backed paper trail supporting return-offer decisions and compliance audits.